GENOMEDL CLINICAL GENOMICS PIPELINE (v3)

Pediatric & Germline Hereditary Cancer Screening Report

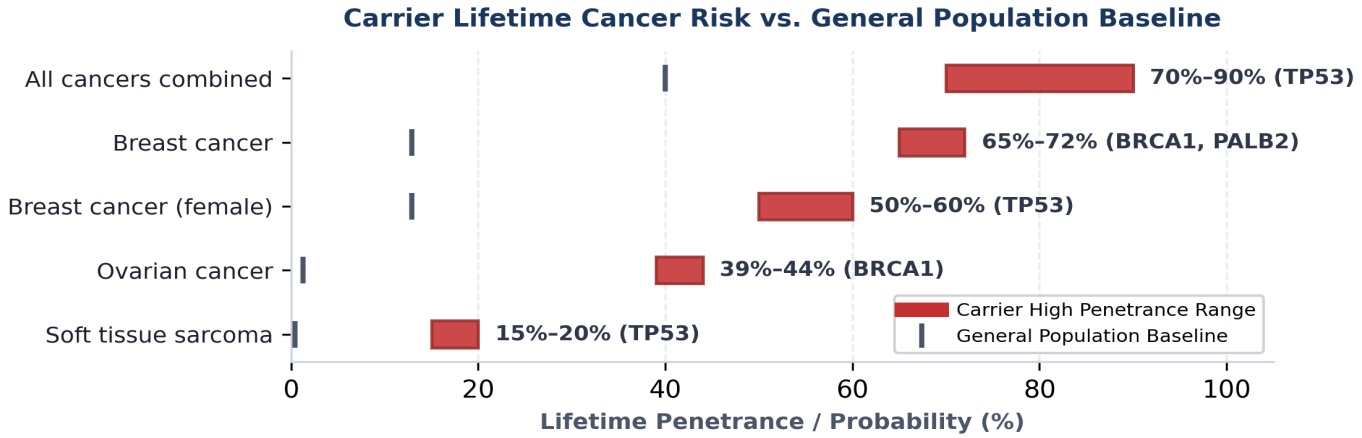
Sample Identifier:	SAMPLE_DEMO	**Reference Genome:**	GRCh38 / hg38
Panel Profile:	130 Hereditary Cancer Genes	**Analysis Date:**	November 07, 2026 (18:25 UTC)
Total Variants Filtered:	5 candidates	**Actionable Tiers:**	**3 Pathogenic / Likely Pathogenic**

ACTIONABLE CARRIER STATUS IDENTIFIED (3 Variants)

High-penetrance pathogenic germline variants were identified in hereditary cancer predisposition genes. Published lifetime carrier risks and clinical surveillance recommendations are summarized below.

1. Executive Summary: Lifetime Cancer Risk by Organ / Type

CANCER TYPE / ORGAN	CARRIER RISK RANGE	GENE(S) INVOLVED	AGE HORIZON	POPULATION BASELINE
All cancers combined	**70% - 90%**	TP53	Up to age 70	~40.0% general pop.
Breast cancer	**65% - 72%**	BRCA1, PALB2	Up to age 80	~12.9% general pop.
Breast cancer (female)	**50% - 60%**	TP53	Up to age 60	~12.9% general pop.
Ovarian cancer	**39% - 44%**	BRCA1	Up to age 80	~1.3% general pop.
Soft tissue sarcoma	**15% - 20%**	TP53	Up to age 60	~0.4% general pop.



2. Detailed Genomic Variant Classifications

Each candidate variant was filtered against gnomAD population frequencies (AF > 1% = Benign), matched against ClinVar curated submissions (≥2 stars), and unclassified novel variants were scored with the GenomeDL v3 ESM-augmented Multimodal Fusion ResNet.

GENE & LOCUS	HGVS / PROTEIN	GT (DP)	CLASSIFICATION	EVIDENCE SOURCE	MODEL P
BRCA1 17:43106487 A>C	Curated clinical consensus from ClinVar submission (Variation ID 17661)	0/1 (Het)	**Pathogenic**	ClinVar lookup (3 stars: reviewed by expert panel)	0.999
TP53 17:7675088 C>T	Curated clinical consensus from ClinVar submission (Variation ID 12374)	0/1 (Het)	**Pathogenic**	ClinVar lookup (3 stars: reviewed by expert panel)	0.999
BRCA2 13:32338780 A>G	High population frequency in gnomAD indicates benign polymorphism	0/1 (Het)	**Benign**	Common polymorphism (gnomAD AF = 4.50% > 1.0%)	0.001
MUTYH 1:45798000 C>T	Calibrated probability P(pathogenic) = 0	0/1 (Het)	**Carrier (Recessive)**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.149
PALB2 16:23635338 C>T	Calibrated probability P(pathogenic) = 0	0/1 (Het)	**Likely Pathogenic**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.851

3. Actionable Variant Findings & Evidence Highlights

BRCA1 (chr17:43106487 A>C) — PATHOGENIC | Source: ClinVar lookup (3 stars: reviewed by expert panel) | Model P: 0.9990

• **Breast cancer**: 65%–72% (up to age 80) [Ref: Kuchenbaecker et al. JAMA 2017]; • **Ovarian cancer**: 39%–44% (up to age 80) [Ref: Kuchenbaecker et al. JAMA 2017]

TP53 (chr17:7675088 C>T) — PATHOGENIC | Source: ClinVar lookup (3 stars: reviewed by expert panel) | Model P: 0.9990

• **All cancers combined**: 70%–90% (up to age 70) [Ref: Chompret et al. / NCCN 2024]; • **Breast cancer (female)**: 50%–60% (up to age 60) [Ref: Olivier et al. 2010]; • **Soft tissue sarcoma**: 15%–20% (up to age 60) [Ref: Olivier et al. 2010]

PALB2 (chr16:23635338 C>T) — LIKELY PATHOGENIC | Source: GenomeDL v3 ESM-Augmented Fusion ResNet | Model P: 0.8507

• **Breast cancer**: 40%–60% (up to age 80) [Ref: Antoniou et al. NEJM 2014]

4. Computational Methodology & Validation Standards

Model Architecture: Multimodal Fusion ResNet combining 26 tabular biochemical features, syndrome embeddings, and ESM-2 (t12_35M_UR50D) masked-marginal log-likelihood ratios and hidden states.
Calibration & Benchmark: Validation-fitted Temperature Scaling (T=0.9357, ECE=0.0249). On 5-fold Gene-Held-Out cross-validation (GroupKFold), model achieves ROC-AUC **0.9042 [95% CI: 0.8872 – 0.9168]** on unseen genes vs tree baseline 0.7225.

MANDATORY CLINICAL GENOMICS DISCLAIMER:
Research prototype, not a diagnosis. Confirm in a certified clinical diagnostic laboratory with formal genetic counselling. Testing children or minors for adult-onset conditions requires specialized pediatric genetic counselling and informed consent.